

Solution

In essence, ‘feature evaluation’ (measuring ‘feature effectiveness’) is a crucial step in a successful software product development life cycle. It addresses the array of challenges and complications outlined above by systematically assessing the efficacy of deployed features and taking necessary actions based on the gained insights. Feature effectiveness can be measured in two ways: ‘Quantitative indicators’ (usage metrics, conversion rates, retention rates, click-through rates, error rates, etc) and ‘qualitative feedback’ (user surveys, usability testing, A/B testing, support data, user review, etc). Quantitative data helps identify trends and patterns, while qualitative feedback offers context and deeper insights. Combining quantitative indicators and qualitative feedback provides a comprehensive picture of feature effectiveness.

Quantitative indicators can be grouped into four

Daily active users (DAU): This category of quantitative indicators focuses on the frequency and engagement of users with a product or service. DAU measures how many users interact with a platform or application daily. It provides a real-time snapshot of user activity, indicating the level of interest and dependence on the product. A high DAU suggests that a product is effectively retaining its user base and driving consistent engagement.

Session duration (SD): The session duration metrics delve into the depth of user engagement. SD tracks the average amount of time a user spends within a single session or interaction with the product. A longer session duration often indicates that users find value and contentment in their interactions, which can be a sign of a well-designed and engaging product.

Conversion rate (CR): Conversion rate metrics are crucial for understanding how effectively a product is at converting potential customers into paying customers. Whether it’s signing up for a newsletter, making a purchase, or taking any desired action, CR quantifies the success of these conversion events. A high CR signifies that the product or marketing strategy is effective in persuading users to take the desired actions.

Customer lifetime value (CLV): CLV is a comprehensive indicator that analyses the long-term profitability of customers. It quantifies the total value a customer brings to the business over their entire relationship with the company. By considering factors such as customer retention, repeat purchases, and the value of referrals, CLV offers insights into the sustainability and growth potential of a business.

All other quantitative indicators can be derived from these four major categories — e.g., monthly active users from DAU; retention rates from a combination of DAU, SD, CR and CLV; click-through rates from CR and CLV; engagement metrics from DAU, SD and CR; revenue metrics from the CLV, etc.

The method for tracking and measuring the effectiveness of the deployed user features in a production environment can be a three-step process:

- Calculating the mathematical notation or numerical representations (FEM or feature effectiveness measure, and FEI or feature effective indicators) that can clearly and quantitatively represent the effectiveness of a feature using the four categories listed earlier.
- Feed the FEM/FEI into the source and analytical systems.
- Analyse if the deployed features are yielding the desired results.

Calculating feature effectiveness measure (FEM) and relative deviations

▪ *Category weightage (CWT)*

Assigning CWT is a crucial step in creating a structured evaluation framework for various features, metrics, or aspects of a product, service, or business. CWT represents the significance or importance that each category carries within the assessment process. The total CWT should always sum up to 100 to maintain a balanced and comprehensive evaluation. However, these weightages can be adjusted based on the specific characteristics and objectives of the features being analysed.

- *Importance of CWT:* CWT allows decision-makers to prioritise and emphasise certain aspects of a product or business over others, aligning them with strategic goals. It helps create a clear hierarchy of importance within an evaluation framework, ensuring that critical areas receive the appropriate attention.
- *Customisation for different features:* Not all features, metrics, or business aspects are equal in importance. The CWT may differ from one feature to another based on their role and relevance. Features with varying objectives or characteristics may require unique CWT assignments. For instance, a high-margin, low-volume feature might give more weight to profitability (e.g., CLV), while a growth-focused feature could prioritise user engagement (e.g., DAU).
- *Summation to 100:* The total CWT should always equal 100 to ensure that the evaluation remains balanced and comprehensive. This constraint reflects the principle that the entire assessment is a holistic view of a product’s or business’s performance. The 100-point scale simplifies comparisons and decision-making by offering a clear, standardised basis for assessment.
- *Dynamic adjustments:* CWT is not fixed and can evolve over time. As business goals change, or as features or strategies are adjusted, the CWT can be modified to align with the new priorities. Data-driven approaches, such as historical performance data and market dynamics, can inform these adjustments.
- *Flexibility and iteration:* Assigning CWT is not a one-time process but an ongoing and iterative one. It’s essential to regularly review and adapt the weightings to keep the evaluation framework